DGMS 2019 • SECTION I: HYDROGEOLOGY & ENVIRONMENTAL GEOLOGY

Question 70: Passive Microwave Remote Sensing & Dielectric Controls on Hydrogeological Mapping

Assertion-Reason Analysis, Radiometric Brightness Temperature, and Soil Permittivity Dynamics

OFFICIAL EXAMINATION QUESTION

Consider the following statements:

Statement 1: Passive microwave remote sensing can help in detecting seepage zones, springs, faults, soil moisture variations, etc.

Statement 2: Water and vegetation have dielectric constants that are very different from those of other ground objects.

Which one of the following is correct in respect of the above statements?

(a) Both Statement 1 and Statement 2 are true and Statement 2 is the correct explanation of Statement 1 [CORRECT]

(b) Both Statement 1 and Statement 2 are true and Statement 2 is not the correct explanation of Statement 1

(c) Statement 1 is true but Statement 2 is false

(d) Statement 1 is false but Statement 2 is true

VERIFIED ANSWER KEY

Correct Option: (a) Both Statement 1 and Statement 2 are true and Statement 2 is the correct explanation of Statement 1 — In passive microwave radiometry, the emitted radiant power is governed by the *dielectric constant* (ε) of the medium. Liquid water has an exceptionally high dielectric constant ($\varepsilon \approx 80$) compared to dry silicate rocks and dry soils ($\varepsilon \approx 3\text{--}5$). Vegetation moisture also exhibits high permittivity ($\varepsilon \approx 30\text{--}50$). Because changes in water

content induce massive shifts in microwave emissivity and brightness temperature, passive microwave sensors can readily map seepage zones, groundwater discharge, fault traces, and soil moisture patterns. Statement 2 is the direct physical cause of Statement 1.

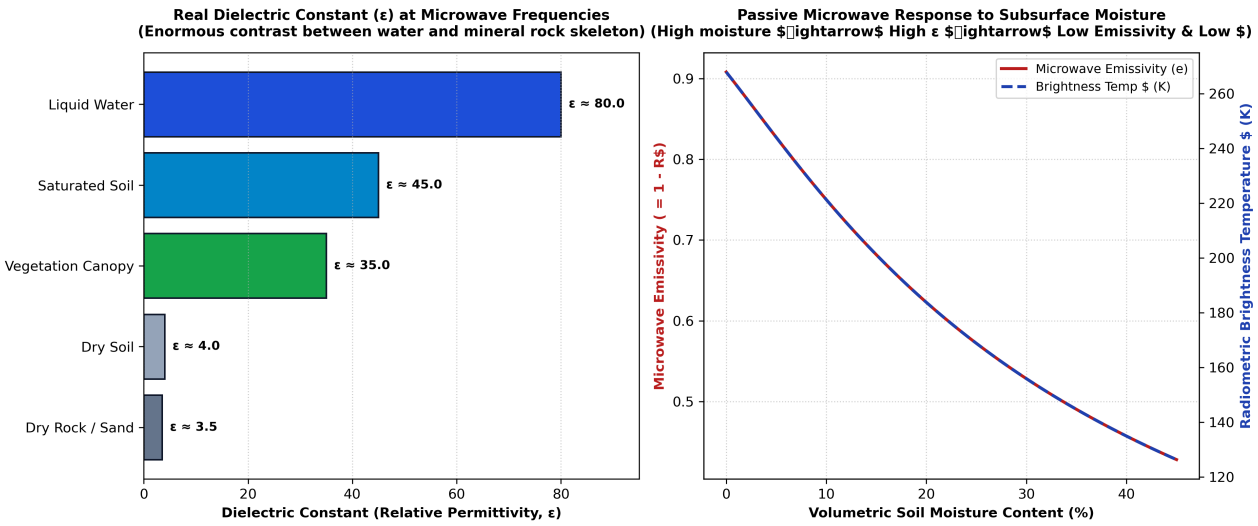


Figure 70.1 | Physical Basis of Microwave Moisture Sensing: (Left) Giant contrast in real dielectric constant (ε) at microwave frequencies (1.4 GHz L-band) between dry mineral soils/rocks ($\varepsilon \approx 3\text{--}4$) and free liquid water ($\varepsilon \approx 80$). (Right) As volumetric soil moisture increases, surface reflectivity rises sharply, causing microwave emissivity ($e = 1 - R$) and radiometric brightness temperature ($T_B = e \cdot T_{\text{phys}}$) to drop from ~280 K down to ~150 K, providing a sensitive diagnostic signature for subsurface seepage and fault springs.

1. Physical Principles of Passive Microwave Radiometry

All natural terrestrial objects above absolute zero emit thermal radiation across the electromagnetic spectrum. In the microwave region (wavelengths $\lambda \approx 1\text{ mm}$ to 1 m; frequencies $0.3\text{ to }300\text{ GHz}$), the Rayleigh-Jeans approximation to Planck's radiation law applies:

$$T_B = e \cdot T$$

where:

- T_B is the **Brightness Temperature** (the equivalent blackbody temperature detected by the passive microwave radiometer antenna in Kelvins).
- T is the actual **Physical / Thermodynamic Temperature** of the surface in Kelvins (~270 to 300 K).
- e is the **Microwave Emissivity** of the terrestrial material ($0 \le e \le 1$).

According to Kirchhoff's radiation law for an opaque medium, emissivity is inversely linked to surface power reflectivity (R):

$$e = 1 - R$$

2. The Crucial Role of the Dielectric Constant (ε)

At microwave frequencies, the Fresnel reflectivity (R) at normal incidence is determined by the complex relative permittivity (**dielectric constant**, ε):

$$R = \left| \frac{\sqrt{\varepsilon} - 1}{\sqrt{\varepsilon} + 1} \right|^2$$

The dielectric constant is a measure of an electric dipole's ability to polarize in response to an oscillating electromagnetic field:

- **Dry Silicate Minerals & Dry Rocks:** Tightly bound electrons; polarization is weak $\rightarrow$ $\varepsilon_{\text{dry}} \approx 3\text{ to }5$. Emissivity is very high: $e \approx 0.90\text{ to } 0.95$. Brightness temperature is warm: $T_B \approx 270\text{ K}$.
- **Pure Liquid Water:** Water molecules ($\mathrm{H_2O}$) possess permanent, highly mobile electric dipoles that rotate freely at low microwave frequencies (L-band, 1.4 GHz) $\rightarrow$ $\varepsilon_{\text{water}} \approx 80$. Reflectivity is huge ($R \approx 0.60$), causing emissivity to plummet: $e \approx 0.40$. Brightness temperature is cold: $T_B \approx 120\text{ K}$.
- **Vegetation Canopy:** Contains high moisture in cellular sap and leaves $\rightarrow$ $\varepsilon_{\text{veg}} \approx 20\text{ to }50$, creating distinct microwave depolarization and attenuation signatures.

3. Geological & Hydrogeological Applications (Statement 1 Verification)

Because adding even 5% to 15% moisture to a dry soil shifts its dielectric constant from 4 to over 15, the radiometric brightness temperature drops by 40 to 80 Kelvins! This physics makes passive microwave remote sensing uniquely powerful for:

- **Detecting Seepage Zones & Springs:** Saturated ground surrounding springs appears as sharply defined cold radiometric anomalies ($T_B < 200\text{ K}$) against dry ambient terrain ($T_B \approx 270\text{ K}$).
- **Mapping Fault Zones:** Fault gouge, breccia zones, and fractured damage zones act as conduits or barriers to groundwater, concentrating shallow moisture along structural lineaments.
- **Soil Moisture Variations:** Dedicated satellite missions like SMOS (Soil Moisture and Ocean Salinity) and SMAP (Soil Moisture Active Passive) operate at L-band (1.4 GHz) to map global upper-soil moisture accurately to within 4% volumetric water content.

- **Buried Paleochannels:** Subsurface sandy channels retaining perched groundwater under dry desert sands can be traced due to low microwave emissivity.

4. Assertion-Reason Evaluation

Statement	Content	Scientific Status	Logical Relationship
Statement 1	Passive microwave remote sensing can help in detecting seepage zones, springs, faults, soil moisture variations, etc.	**True**	Observation / practical hydrogeological remote sensing capability.
Statement 2	Water and vegetation have dielectric constants that are very different from those of other ground objects.	**True**	**Direct physical explanation!** The $\sim$80 vs. 3–5 dielectric contrast is the exact physical mechanism enabling Statement 1.

5. Authoritative Textbook References

- **Thomas M. Lillesand, Ralph W. Kiefer & Jonathan W. Chipman (2015)** *Remote Sensing and Image Interpretation*, 7th Ed., John Wiley & Sons, Chapter 6 ("Microwave and LiDAR Sensing: Passive Microwave Radiometry & Dielectric Properties"), pp. 385–410.
- **Fawwaz T. Ulaby, Richard K. Moore & Adrian K. Fung (1986)** *Microwave Remote Sensing: Active and Passive*, Vol. III: From Theory to Applications, Artech House, Chapter 19 ("Passive Microwave Sensing of Soil Moisture").
- **B. Bhatta (2021)** *Remote Sensing and GIS*, 3rd Ed., Oxford University Press, Chapter 7 ("Microwave Remote Sensing").
- **Floyd F. Sabins (2007)** *Remote Sensing: Principles and Interpretation*, 3rd Ed., Waveland Press, Chapter 6 ("Radar and Microwave Remote Sensing").